REVERSE ENGINEERING : SMARTPHONE

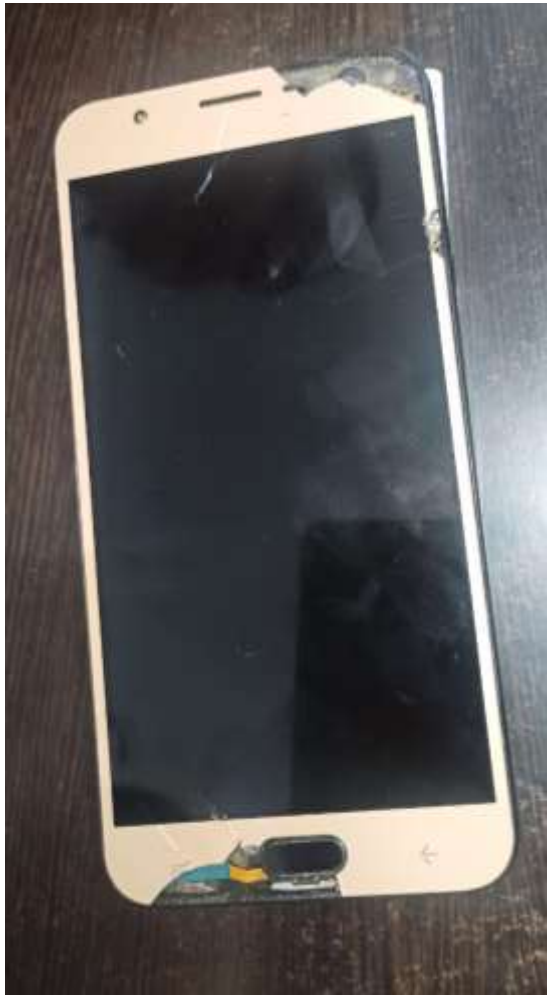
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EXTERNAL PRODUCT INSPECTION

Samsung Galaxy J7 Duo

- Manufacturer: Samsung
- Smartphone with 5.5" display
- Physical buttons and touchscreen
- Rear dual-camera system
- Micro-USB
- Removable rear cover and battery





PRODUCT LITERATURE & VERSION REVIEW

Parameter	Specification
Processor	Exynos 7885
CPU	Octa-core
GPU	Mali-G71
RAM	4 GB
Storage	32 GB
Display	5.5" HD AMOLED
Battery	3000 mAh
OS at launch	Android 8.0 Oreo
Connectivity	2G / 3G / 4G LTE

PURPOSE, USER NEEDS & TRADE-OFFS

- Combines **communication, internet, photography and computing** in one portable device.
- Supports calls, messaging, applications and multimedia.

User Expectations

- Good battery life
- Reliable performance
- Good display
- Good camera
- Sufficient storage
- Reliable connectivity
- Easy operation

Engineering Trade-offs –

- **Performance ↔ Battery consumption**
- **Large display ↔ Portability**
- **Features ↔ Cost**
- **Battery capacity ↔ Weight/thickness**
- **Camera system ↔ Internal space**

TEARDOWN PROCESS & SAFETY

Tools

- Precision screwdriver set
- Plastic pry tools
- Tweezers
- ESD wrist strap
- ESD mat
- Magnifying glass
- Camera/phone
- Screw containers
- Labels/markers

Safety

- Powering off the phone
- Removing the battery before PCB work
- Using ESD protection
- Holding PCBs by their edges
- Avoiding touching IC pins
- Keeping screws organized



MECHANICAL DISASSEMBLY

External Mechanical Parts

- Removable rear cover
- Removable 3000 mAh battery
- Front display assembly
- Physical Home button
- Power button
- Volume buttons
- SIM card slots
- microSD slot
- Micro-USB port
- 3.5 mm audio jack
- Speaker openings
- Camera openings

Internal Mechanical Features

- PCB mounting points
- Screws
- Plastic clips
- Battery contacts
- Camera modules
- Antenna structures
- Display connectors





COMPONENT INVENTORY

Component	Function
Main PCB	Contains electronic circuits
Exynos 7885 SoC	Main processing
RAM	Temporary data storage
Flash storage	Permanent data storage
Battery	Power source
Display	Visual output
Rear cameras	Image capture
Front camera	Selfies/video
Fingerprint sensor	Biometric security
Accelerometer	Motion detection
Speaker	Audio output
Microphone	Audio input
USB connector	Charging/data
Antennas	Wireless communication
Power-management ICs	Voltage/power regulation



COMPONENT IDENTIFICATION & SPECIFICATIONS

Main Processor

Exynos 7885

- Manufacturer: Samsung
- CPU: Octa-core
- GPU: Mali-G71
- Used as the main SoC

Memory

- RAM: **4 GB**
- Internal storage: **32 GB**
- Expandable storage: **microSD up to 256 GB**

Display

- Type: Super AMOLED/HD AMOLED
- Size: 5.5 inches
- Resolution: 1280 × 720

COMPONENT FUNCTIONAL SPECIFICATIONS

Processor

Function: Executes instructions and controls system operation.

RAM

Function: Temporarily stores data used by running applications.

Flash Memory

Function: Stores Android, applications and user files.

Battery

Specification: 3000 mAh Li-ion; removable

Function: Provides electrical power.

Cameras

Rear: 13 MP + 5 MP depth camera

Front: 8 MP

Function: Image/video capture.

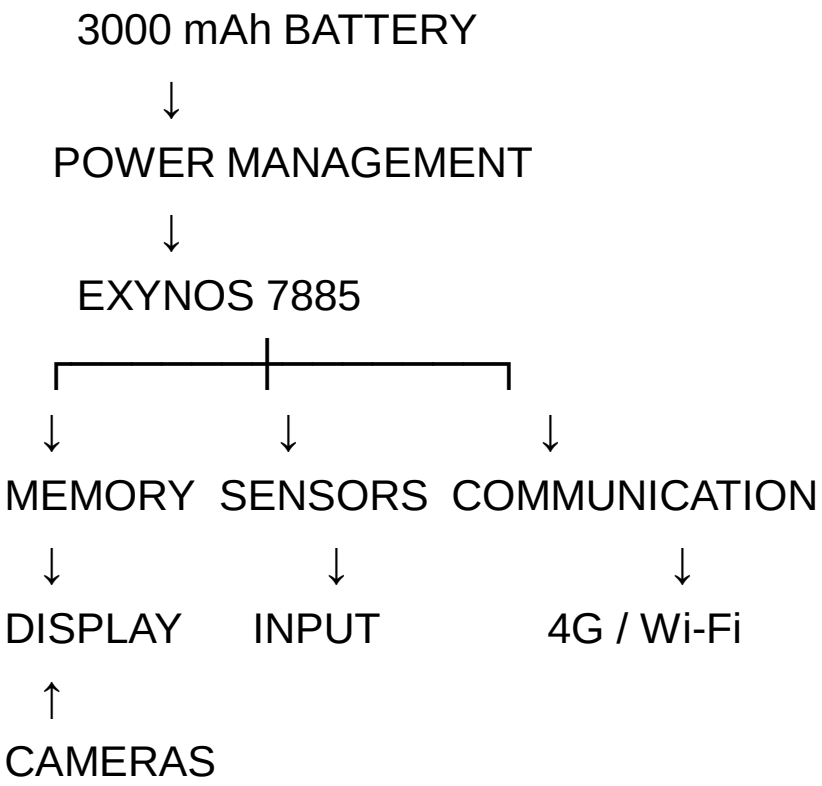
Sensors

- Accelerometer
- Proximity sensor

Display

Function: Provides visual information and touchscreen interaction.

SYSTEM ARCHITECTURE



INPUT–PROCESS–OUTPUT ANALYSIS

Inputs

- Touchscreen
- Physical buttons
- Camera
- Microphone
- Fingerprint sensor
- Light sensor
- Cellular signals
- Wi-Fi signals
- Bluetooth signals

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Processing

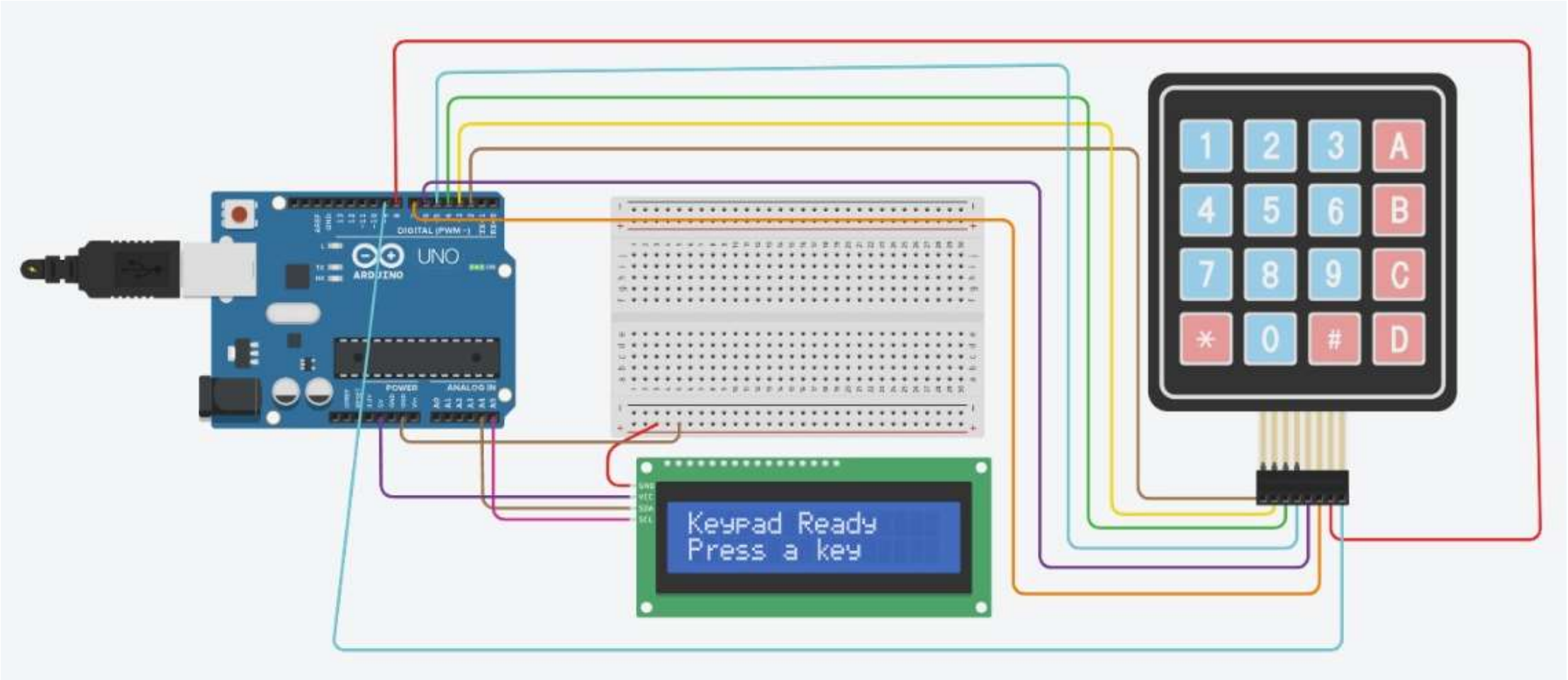
- Exynos processor + RAM + operating system + supporting ICs

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Outputs

- AMOLED display
- Speaker
- Vibration motor
- Cellular communication
- Wi-Fi communication
- Bluetooth communication
- Camera storage/results

TINKCAD SIMULATION



FINAL REVERSE-ENGINEERING CONCLUSIONS

The Samsung Galaxy J7 Duo is a compact smartphone that combines several engineering systems into one portable product. Its major subsystems include the battery and power-management system, processor, memory, display, camera modules, sensors, audio system and wireless communication system.

The external inspection shows that the phone was designed to provide multiple functions while maintaining a relatively compact form factor. Its 5.5-inch Super AMOLED display provides the primary visual interface, while the processor and memory system perform the required computational tasks. The dual rear-camera system provides enhanced photography capabilities, while sensors and the fingerprint system provide additional functionality.

The reverse-engineering process demonstrates that the phone is not simply a collection of individual components. Each component has an electrical and functional purpose, and the components must work together as a complete system.

The major engineering trade-offs include **performance versus power consumption, battery capacity versus size and weight, camera capability versus internal space, and features versus manufacturing cost**.

Therefore, studying the Galaxy J7 Duo through reverse engineering helps reveal the engineering decisions involved in creating a consumer electronic product and demonstrates how mechanical, electrical and software systems are integrated into a single device.

THANK YOU